

AKSHAYA VERMA

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student (Cyber Security) with hands-on experience designing REST APIs and backend services with Node.js, Express, PostgreSQL, and Prisma. Built and deployed a full-stack platform with authentication, access control, and hardened APIs, plus a server-side execution service with process isolation and resource limits. Seeking a Full Stack / Backend Engineering role.

EDUCATION

VIT University, Bhopal

B.Tech in Computer Science Engineering (Cyber Security)

2023 -- 2027

CGPA: 8.01 / 10.00

Relevant Coursework: DBMS, Operating Systems, Computer Networks, Software Engineering, Data Privacy, Software Vulnerability Testing.

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript

Backend & APIs: Node.js, Express.js, REST APIs, JWT, RBAC, Prisma ORM, Rate Limiting, Security Headers

Frontend: React.js, Tailwind CSS

Databases: PostgreSQL, SQLite, Firebase

Cloud & Tools: AWS (EC2, S3, RDS, IAM, VPC), Linux, Git, GitHub, Postman

AI / GenAI: LLM APIs, RAG, Embeddings

TECHNICAL PROJECTS

CyberSec RAG -- AI-Powered Cybersecurity Knowledge Assistant | *Backend*

In Development

Node.js, Express, LLM APIs, RAG, Vector DB

- **Building** a Node.js/Express pipeline that processes documents into chunks and generates embeddings for each chunk.
- **Storing** embeddings in a vector database for semantic retrieval against incoming natural-language queries.
- **Designed** a retrieve-then-generate API flow that passes relevant context to an LLM so answers stay grounded in source documents.
- **Evaluating** prompt injection and over-permissioned tool calling while integrating LLM APIs and tool calling into the backend.

Persona -- Privacy-Focused Anonymous Community Platform | *Full Stack*

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Node.js, Express, PostgreSQL, Prisma, React, JWT

- **Built and deployed** a full-stack community platform with a React frontend and Node.js/Express REST API backend.
- **Modeled** posts, threaded discussions, reactions, and trust scoring with PostgreSQL and Prisma.
- **Implemented** JWT authentication and authorization checks to protect user identity and restrict access to resources.
- **Hardened** the API with rate limiting, Helmet security headers, and Prisma field-level selection to limit abuse and data exposure.
- **Added** server-side content validation that detects emails, phone numbers, and personal links before persistence.

BreakCase -- Secure Code Execution & Counterexample Testing Platform | *Full Stack*

React, TypeScript, Node.js, Express, Prisma, C++17

- **Built** a full-stack platform with a React and TypeScript frontend and a Node.js/Express backend that generates counterexamples to expose incorrect logic in C++17 solutions.
- **Developed** a server-side service that compiles submitted programs and evaluates generated inputs through controlled Node.js child processes.
- **Enforced** execution timeouts and output limits to prevent runaway execution and uncontrolled output.
- **Validated** inputs and applied resource constraints to test submissions safely against malformed, excessive, and adversarial inputs.

ACHIEVEMENTS

- **Health Hackathon -- Finalist:** Reached the final round among 100+ teams.
- **Competitive Programming:** 25+ contests on CodeChef and Codeforces.